

CCT College Dublin Continuous Assessment

Programme Title:	HDS DAB in Statistics in Data Analytics		
Delivery Mode:	FT/ PT		
Cohort Details:	<i>HDS DAB Sep26 PT SB/PT, Semester 1</i>		
Module Title(s):	<i>Statistical Techniques for DA</i> <i>Programme schedules are all published on the CCT IQR Provider Profile</i>		
Assignment Type:	<i>Individual</i>	Weighting(s):	40%
Assignment Title:	<i>Individual CA1 -</i>		
Lecturer(s):	<i>Cristiani Eccher</i>		
Issue Date:	<i>24th September 2026</i>		
Submission Deadline Date:	<i>Sunday 1st November 2026@11:59 PM</i>		
Late Submission Penalty:	Late submissions will be accepted up to 5 calendar days after the deadline. All late submissions are subject to a penalty of 10% of the mark awarded. Submissions received more than 5 calendar days after the deadline above will not be accepted and a mark of 0% will be awarded.		
Method of Submission:	This assignment is submitted via Moodle.		
Instructions for Submission:	1. Dataset Extracted - Survey 2. Word Document 1000 words and graphs using references, intro, conclusion, structure. 3. Provide the Jupyter Notebook file. Upload all files separately, such as MS word file, jupyter notebook, additional dataset/datasets and any supporting information on Moodle. No pdf and zip files will be accepted.		
Feedback Method:	Results posted in Moodle gradebook		

Assessment Outline

Assessment Task

You are required to collect, process, analyse and interpret the data in order to identify and recommend possible solutions for a specific problem. (Examples like: house energy consumption, time for exercises, healthy food, etc..) using descriptive statistics.

Data source: You will create a survey using Google Forms (or any other tool) for data collection with the following requirements.

1. Add one description in your survey for data protection compliance. Make explicit that the data is for academic purpose and the data will be anonymised. By answering the survey the person is aware of the condition.
2. Columns (Questions) Minimum 10 columns/questions • Data Type: minimum 3 data types (integer, continuous, categories – Likert scale, Boolean) • Rows: minimum 30 observations.
3. Save this data source as “Data_CA1.csv” – **Anonimise your data source by removing any reference to Names, IDs, Address.**

Statistical Analysis

You will perform the analysis of each column and potential relationship of columns. In addition use histograms, box-plots to demonstrate valuable insights related to your data collected and processed. Minimum 8 explorations/graphs related to your analysis.

Report

You will create a report with the following structure:

1. Introduction: motivation why did you choose the referred topic for data collection
2. Analysis/Insights: describe the data explored in your statistical Analysis and explain why this information is relevant
3. Conclusion and Discussion: Conclude your work with your findings, what did you discover? What type of recommendation could be proposed? Approximate 1000 words +/- 200

Assessment Details:

Statistical Techniques for Data Analytics

1. Descriptive Analysis: Summarise your dataset clearly, using relevant descriptive statistics. These should be carefully motivated and justified, and clearly presented. You should critically analyse your findings, in addition to including the necessary Python code, output and plots in the report. [0-30]
2. Use minimum 8 plots and appropriate visualisations/tables or summaries. You must explain your reasoning and the techniques you have used. You must work with Python and your mathematical reasoning must be documented in your report. [0-30]
3. Explore outliers of your data and compare to the previous dataset (with outliers) what conclusions that you observe? [0-10].
4. Inferential Statistics: Use probability (posterior and prior) and Bayes theory for a specific feature and show the results. Also using your database normalised, perform a probability test, such as what is the probability of event A being given event B? Using p-norm for that. [0-15%].
5. Correlation/Chi-Square Analysis: Investigate correlations and relationship of features and target value available in your dataset to identify potential predictors [0-15%].

Assessment Requirements

All assessment submissions must meet the following minimum requirements:

- All files (MS word, Jupyter notebook, data files and any supporting document) should be uploaded separately on Moodle.
- The rubric is provided for the detailed breakdown of marks at the end of this integrated CA2 for all modules.
- Number of Words in the report (100 words +/-20%) excluding diagrams, code, references, citations and titles.
- Submission the report in Word format and. IPYNB file containing the code. The name of the file should be Hdip_YOUR_NAME_Year.
- Be submitted by the deadline date specified or be subject to late submission penalties.
- Use [Harvard Referencing](#) when citing third party material.
- Be the student's own work.
- Include the CCT assessment cover page.
- Read carefully the Artificial Intelligence policy (GenAI) policy for this integrated assignment.

Use of Artificial Intelligence

Artificial Intelligence (AI) can be used in this assignment in the following way(s):

- Brainstorming and refining your ideas;
- Fine tuning your research questions;
- Finding information on your topic;
- Drafting an outline to organise your thoughts;
- Checking grammar and style
- Other

Note for students:

- You are required to include all AI prompt(s) and outputs in an appendix, including a short reflection (approx. 250 words) on how AI enhanced your work, or otherwise. These words are not part of 1000 words asked for CA1.
- You are responsible for the information you submit based on an AI query. Your use of AI tools must be properly documented and cited.
- Any assignment that is found to have used generative AI tools in an unauthorised way will be subject to college disciplinary procedures as outlined in the QA Manual.
- Cite all AI references used in your assignment.
- You must declare how you used AI on your assignment cover sheet.
- In the case of any suspicious work submitted by a student, the student will be called for a Q&A session.
- When in doubt about permitted usage, please ask for clarification to your lecture.

Statistical Techniques for DA

Marking Criteria	Weighting	Excellent (+70%)	Very Good (60-69%)	Good (50-59%)	Acceptable (40-49%)	Fail (<39%)
Descriptive Analysis	0-30	Thoroughly explore the dataset's characteristics, normality, skewness, outliers and central tendency theory	Accurately describes normality, skewness, outliers and central tendency theor	Provides a good overview of dataset characteristics with some relevant details about normality and skewness	Basic dataset description provided, lacking key details in one or more areas regarding normality and skewness.	Insufficient or incorrect description of the dataset characteristics regarding normality and skewness
Statistical visualizations	0-30	Thoroughly demonstrates the distribution of the data using appropriate visualisation	Accurately demonstrates the distribution of the data using appropriate visualisations	Provides a good overview demonstrating the distribution of the data using appropriate visualisations	Basic dataset description provided, demonstrating the distribution of the data using appropriate visualisations	Insufficient or incorrect description of the dataset demonstrating the distribution of the data using appropriate visualisations
Outliers and statistical analysis	0-10	Clear, comprehensive application of sampling and confidence interval	Strong use of application of sampling and confidence interval.	Uses good use of application of sampling and confidence interval.	Basic application of use of application of sampling and confidence interval.	Limited or no use use of application of sampling and confidence interval.
Inferential Statistics	0-15	Comprehensive application of Inferential Statistics: Use probability (posterior and prior) and bayes theory for a specific feature and show the results.	Strong application of Inferential Statistics: Use probability (posterior and prior) and bayes theory for a specific feature and show the results.	Good application of Inferential Statistics: Use probability (posterior and prior) and bayes theory for a specific feature and show the results.	Limited Application of Inferential Statistics: Use probability (posterior and prior) and bayes theory for a specific feature and show the results.	Minimal or incorrect application Inferential Statistics: Use probability (posterior and prior) and bayes theory for a specific feature and show the results.
Correlation/Chi-Square	0-15	Comprehensive application of correlation/chi-square and relationships between features and target value	Strong application of correlation/chi-square and relationships between features and target value.	Good application of correlation/chi-square and relationships between features and target value.	Limited Application of correlation/chi-square and relationships between features and target value.	Minimal or incorrect application of correlation/chi-square and between features and target value.

The Irish Grading System

The grading system in CCT is the QQI percentage grading system and is in common use in higher education institutions in Ireland. The pass mark and thresholds for different grade bands may be different from what you have experienced in the higher education system in other countries. CCT grades must be considered in the context of the grading system in Irish higher education and not assumed to represent the same standard the percentage grade reflects when awarded in an international context.

Additional Information

- Lecturers are not required to review draft assessment submissions. This may be offered at the lecturer's discretion.
- In accordance with CCT policy, feedback to learners may be provided in written, audio or video format and can be provided as individual learner feedback, small group feedback or whole class feedback.
- Results and feedback will only be issued when assessments have been marked and moderated / reviewed by a second examiner.
- Additional feedback may be provided as individual, small group or whole class feedback. Lecturers are not obliged to respond to email requests for additional feedback where this is not the specified process or to respond to further requests for feedback following the additional feedback.
- Following receipt of feedback, where a student believes there has been an error in the marks or feedback received, they should avail of the recheck and review process and should not attempt to get a revised mark / feedback by directly approaching the lecturer. Lecturers are not authorised to amend published marks outside of the recheck and review process or the Board of Examiners process.
- Students are advised that disagreement with an academic judgement is not grounds for review.
- For additional support with academic writing and referencing students are advised to contact the CCT Library Service.
- For additional support with subject matter content students are advised to contact the [CCT Student Mentoring Academy](#)
- For additional support with IT subject content, students are advised to access the [CCT Support Hub](#).